C1 Solutions: functions, graphs and equations

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All answers were checked with SymPy (exact algebra) and, for problem 23, with a numerical count.

Practice

1a. Domain: $8 - 4x \geq 0 \iff x \leq 2$, so $D_f = (-\infty, 2]$. As x runs through $(-\infty, 2]$, $8 - 4x$ runs through all of $[0, \infty)$, so $\sqrt{8-4x}$ takes every value in $[0, \infty)$ and $f(x) = 3 - \sqrt{8-4x}$ takes every value in $(-\infty, 3]$. $B_f = (-\infty, 3]$.

1b. Domain: $x^2 - 9 \neq 0$, so $x \neq \pm 3$. Range: $x^2 - 9$ takes every value in $[-9, 0)$ (for $-3 < x < 3$) and every value in $(0, \infty)$ (for $|x| > 3$).

- If $u \in [-9, 0)$ then $\frac{1}{u} \in (-\infty, -\frac{1}{9}]$ (for example $u = -9$ gives $-\frac{1}{9}$, and $u \to 0^-$ gives $-\infty$).
- If $u \in (0, \infty)$ then $\frac{1}{u} \in (0, \infty)$.

So $B_g = (-\infty, -\frac{1}{9}] \cup (0, \infty)$.

2a. $x \neq -1$.

2b. Vertical asymptote $x = -1$ (the denominator is 0, the numerator is $-5 \neq 0$). Horizontal: $\dfrac{3x-2}{x+1} = \dfrac{3 - 2/x}{1 + 1/x} \to 3$, so $y = 3$.

2c. Write $g(x) = \dfrac{3(x+1) - 5}{x+1} = 3 - \dfrac{5}{x+1}$. The fraction $\frac{5}{x+1}$ takes every value except 0, so g takes every value except 3. $B_g = \mathbb{R} \setminus \{3\}$. (Problem 5 confirms this: the inverse is defined for every $x \neq 3$.)

3. Start with $y = \frac{1}{x}$.

1. Translate $(-1, 0)$: $y = \dfrac{1}{x+1}$.
2. Stretch from the x-axis by factor -3 (stretch by 3 and reflect in the x-axis): $y = -\dfrac{3}{x+1}$.
3. Translate $(0, 2)$: $y = 2 - \dfrac{3}{x+1}$.

(Steps 1 and 2 may be swapped, because one acts inside and the other outside. Step 3 must come after step 2.)

4a. Stretch from the y-axis by $\frac{1}{2}$: replace x by $x/\frac{1}{2} = 2x$: $y = \sqrt{2x}$. Then translate $(3, -1)$: replace x by $x - 3$ and subtract 1: $y = \sqrt{2(x-3)} - 1 = \sqrt{2x-6} - 1$.

4b. Translate first: $y = \sqrt{x-3} - 1$. Then stretch from the y-axis by $\frac{1}{2}$: replace x by $2x$: $y = \sqrt{2x-3} - 1$. This is a different graph: it starts at $(1\frac{1}{2}, -1)$ instead of $(3, -1)$. The horizontal stretch also halves the horizontal translation.

5. $y = \dfrac{3x-2}{x+1} \Rightarrow y(x+1) = 3x - 2 \Rightarrow yx - 3x = -2 - y \Rightarrow x(y-3) = -(y+2) \Rightarrow x = \dfrac{y+2}{3-y}$.

So $g^{-1}(x) = \dfrac{x+2}{3-x}$, domain $x \neq 3$.

6a. $3x+2=8$ or $3x+2=-8$, so $x=2$ or $x=-\frac{10}{3}$.

6b. $|3x+2|\geq 8 \iff 3x+2\geq 8$ or $3x+2\leq -8$, so $x\geq 2$ or $x\leq -\frac{10}{3}$.

7. $x^3-x^2-6x=x(x^2-x-6)=x(x-3)(x+2)$. Zeros $-2,0,3$, each of multiplicity 1, so the sign changes at each zero. For large x the product is positive. Signs from left to right: $-,+,-,+$. **Answer:** $-2<x<0$ or $x>3$.

8a. Square: $2x+9=x^2+6x+9 \Rightarrow x^2+4x=0 \Rightarrow x=0$ or $x=-4$. Check $x=-4$: $\sqrt{1}=1$ but $x+3=-1$. False. Check $x=0$: $\sqrt{9}=3=0+3$. Correct. **Answer:** $x=0$.

8b. Square: $x+5=x^2-2x+1 \Rightarrow x^2-3x-4=0 \Rightarrow x=4$ or $x=-1$. Check $x=-1$: $\sqrt{4}=2$ but $x-1=-2$. False. Check $x=4$: $\sqrt{9}=3=4-1$. Correct. **Answer:** $x=4$.

9a. $2-\dfrac{1}{x^2+1}=\dfrac{2(x^2+1)-1}{x^2+1}=\dfrac{2x^2+1}{x^2+1}=f(x)$.

9b. As $x\to\pm\infty$, $\frac{1}{x^2+1}\to 0$, so the horizontal asymptote is $y=2$. Since $x^2+1\geq 1$, we have $0<\frac{1}{x^2+1}\leq 1$ with the maximum 1 at $x=0$, and every value in $(0,1]$ occurs. So $1\leq f(x)<2$: $B_f=[1,2)$.

10a. $f(-x)=-x^3+x=-(x^3-x)=-f(x)$: odd. **10b.** $g(-x)=x^4+x^2=g(x)$: even. **10c.** $h(1)=2$, $h(-1)=0$. Since $h(-1)\neq h(1)$ it is not even; since $h(-1)\neq -h(1)$ it is not odd. Neither. (One counterexample is enough to disprove a "for all" statement.) **10d.** $k(-x)=\dfrac{-x}{x^2+1}=-k(x)$: odd.

11. $f(x)=(x+3)^2-4$. On $x\leq -3$, f is decreasing and its range is $[-4,\infty)$. $y=(x+3)^2-4 \Rightarrow (x+3)^2=y+4 \Rightarrow x+3=-\sqrt{y+4}$ (negative root, because $x+3\leq 0$). So $f^{-1}(x)=-3-\sqrt{x+4}$, domain $x\geq -4$.

CE-level

12a. Factorise: $x^2-9=(x-3)(x+3)$ and $x^2+x-6=(x+3)(x-2)$. The denominator is zero for $x=-3$ and $x=2$.

- At $x=2$ the numerator is $4-9=-5\neq 0$: vertical asymptote $x=2$.
- At $x=-3$ the numerator is also zero, and for $x\neq -3$ we have $f(x)=\dfrac{x-3}{x-2}$, which is defined at $x=-3$. So $x=-3$ is not an asymptote.

So the graph has exactly one vertical asymptote, $x=2$.

12b. The hole is at $x=-3$, with y-value $\dfrac{-3-3}{-3-2}=\dfrac{6}{5}$. The point $\left(-3,\frac{6}{5}\right)$.

12c. Equal degrees, leading coefficients 1 and 1: $y=1$.

13a. $\sqrt{x+4}=\frac{1}{2}x+2$. Square: $x+4=\frac{1}{4}x^2+2x+4 \Rightarrow \frac{1}{4}x^2+x=0 \Rightarrow x(\frac{1}{4}x+1)=0 \Rightarrow x=0$ or $x=-4$. Check $x=0$: $2=2$. Check $x=-4$: $0=0$. Both are correct.

13b. The domain of f is $x\geq -4$. On $(-4,0)$, test $x=-3$: $f(-3)=1$, $g(-3)=\frac{1}{2}$, so $f>g$. For $x>0$, test $x=5$: $f(5)=3$, $g(5)=4.5$, so $f<g$. At $x=-4$ and $x=0$ they are equal. **Answer:** $-4<x<0$.

14. $\dfrac{1}{x}=-x+p \Rightarrow 1=-x^2+px$ (multiply by $x\neq 0$) $\Rightarrow x^2-px+1=0$. Note that $x=0$ is never a solution of this quadratic, so no solutions are lost or gained. Discriminant $D=p^2-4$.

- $D > 0 \iff p < -2$ or $p > 2$: two intersection points.
- $D = 0 \iff p = \pm 2$: one point, where the line touches the hyperbola (at $x = \frac{p}{2} = \pm 1$, the points $(1, 1)$ and $(-1, -1)$).
- $D < 0 \iff -2 < p < 2$: no intersection points.

15a. $x - 1 \geq 0$, so $D_f = [1, \infty)$. $\sqrt{x-1}$ takes all values in $[0, \infty)$, so $B_f = [1, \infty)$.

15b. $y = 1 + \sqrt{x-1} \Rightarrow \sqrt{x-1} = y - 1$ (requires $y \geq 1$) $\Rightarrow x - 1 = (y-1)^2 \Rightarrow x = (y-1)^2 + 1$. Swap letters: $f^{-1}(x) = (x-1)^2 + 1$ with domain $B_f = [1, \infty)$.

15c. f is increasing, so the intersection points lie on $y = x$ (notes §4). Solve $1 + \sqrt{x-1} = x$: $\sqrt{x-1} = x - 1$. Put $u = \sqrt{x-1} \geq 0$: $u = u^2$, so $u = 0$ or $u = 1$, so $x = 1$ or $x = 2$. Both satisfy the original equation ($1 + 0 = 1$, $1 + 1 = 2$). **Answer:** $(1, 1)$ and $(2, 2)$.

16a. $$x - 1 + \frac{3}{x-2} = \frac{(x-1)(x-2)+3}{x-2} = \frac{x^2 - 3x + 2 + 3}{x-2} = \frac{x^2 - 3x + 5}{x-2}.$$

16b. Vertical asymptote $x = 2$ (numerator at 2 is $3 \neq 0$). As $x \to \pm\infty$, $\frac{3}{x-2} \to 0$, so the graph approaches the oblique asymptote $y = x - 1$. There is no horizontal asymptote.

17. $\dfrac{2x-1}{x+3} - 1 = \dfrac{2x - 1 - (x+3)}{x+3} = \dfrac{x-4}{x+3} \leq 0$. Numerator zero at $x = 4$, denominator zero at $x = -3$. Sign diagram: for $x < -3$ both factors negative, quotient $+$; for $-3 < x < 4$ quotient $-$; for $x > 4$ quotient $+$. **Answer:** $-3 < x \leq 4$ ($x = -3$ excluded because the expression is undefined there, $x = 4$ included because the quotient is 0).

18. *Case 1:* $x^2 - 2x \geq 0$, i.e. $x \leq 0$ or $x \geq 2$. Then $x^2 - 2x = x$, $x^2 - 3x = 0$, $x = 0$ or $x = 3$. Both satisfy the case condition. *Case 2:* $0 < x < 2$. Then $-(x^2 - 2x) = x$, $x^2 - x = 0$, $x = 0$ or $x = 1$. Only $x = 1$ satisfies $0 < x < 2$. Check: $x = 0$: $0 = 0$. $x = 1$: $|-1| = 1$. $x = 3$: $|3| = 3$. **Answer:** $x = 0$, $x = 1$ or $x = 3$.

19a. $g(x) = f\left(\frac{x}{3}\right) - 2 = \frac{x^2}{9} + \frac{4x}{3} - 2$.

19b. *From the formula:* the vertex of $ax^2 + bx + c$ is at $x = -\frac{b}{2a} = -\dfrac{4/3}{2/9} = -6$, and $g(-6) = \frac{36}{9} - 8 - 2 = -6$. Vertex $(-6, -6)$. *By transforming:* $f(x) = (x+2)^2 - 4$ has vertex $(-2, -4)$. Stretch from the y-axis by 3: $(-6, -4)$. Translate $(0, -2)$: $(-6, -6)$. Same answer.

20a. $N(t) = \dfrac{1200}{1 + 6/t}$ for $t > 0$. As $t \to \infty$, $\frac{6}{t} \to 0$, so $N \to 1200$: in the long run about 1.2 million downloads.

20b. Since $t + 6 > 0$ we may multiply without changing the inequality sign: $1200t > 900(t + 6) \iff 300t > 5400 \iff t > 18$. So after 18 weeks.

20c. $N(t+6) = 1200t \Rightarrow Nt + 6N = 1200t \Rightarrow t(1200 - N) = 6N \Rightarrow t = \dfrac{6N}{1200 - N}$, for $0 \leq N < 1200$.

Challenge

21a. $y = \dfrac{ax+b}{cx+d} \Rightarrow y(cx+d) = ax + b \Rightarrow x(cy - a) = b - dy \Rightarrow x = \dfrac{-dy + b}{cy - a}$. So $f^{-1}(x) = \dfrac{-dx + b}{cx - a}$, for $x \neq \frac{a}{c}$.

21b. If $d = -a$, then $f^{-1}(x) = \dfrac{ax+b}{cx-a} = \dfrac{ax+b}{cx+d} = f(x)$. So f is its own inverse. (Its graph is symmetric in the line $y = x$.)

21c. If $ad - bc = 0$, then the numerator is a multiple of the denominator: for example with $c \neq 0$, $ax + b = \frac{a}{c}(cx+d) + \left(b - \frac{ad}{c}\right) = \frac{a}{c}(cx+d) - \frac{ad-bc}{c} = \frac{a}{c}(cx+d)$. So $f(x) = \frac{a}{c}$ for all x in the domain: a constant function, which is not one-to-one and has no inverse. In the computation of 21a, solving $x(cy-a) = b - dy$ fails.

22. Let $t = \sqrt{x-1} \geq 0$, so $x = t^2 + 1$. Then $x + 3 - 4t = t^2 - 4t + 4 = (t-2)^2$ and $x + 8 - 6t = t^2 - 6t + 9 = (t-3)^2$. Since $\sqrt{u^2} = |u|$, the equation becomes $|t-2| + |t-3| = 1$. The left side is the sum of the distances from t to 2 and to 3 on the number line. If $2 \leq t \leq 3$, that sum is exactly $3 - 2 = 1$. If $t < 2$ or $t > 3$, the sum is more than 1. So $2 \leq t \leq 3$, i.e. $4 \leq x - 1 \leq 9$: **Answer:** $5 \leq x \leq 10$ (a whole interval of solutions). Check $x = 7$: $t = \sqrt{6}$, $|\sqrt{6} - 2| + |\sqrt{6} - 3| = (\sqrt{6} - 2) + (3 - \sqrt{6}) = 1$.

23. Let $h(x) = x^2 - 4|x| + 3$. It is even, so first study $x \geq 0$: $h(x) = x^2 - 4x + 3 = (x-1)(x-3) = (x-2)^2 - 1$. On $x \geq 0$: $h(0) = 3$, zeros at 1 and 3, minimum -1 at $x = 2$. So $g = |h|$ on $x \geq 0$: decreases from 3 to 0 on $[0,1]$, increases to 1 on $[1,2]$, decreases to 0 on $[2,3]$, then increases to ∞. The graph for $x < 0$ is the mirror image.

Count solutions of $g(x) = k$ for $x > 0$ and double, then treat $x = 0$ separately ($g(0) = 3$).

k	solutions with $x > 0$	total
$k < 0$	0	0
$k = 0$	2 ($x = 1, 3$)	4
$0 < k < 1$	4 (one in each of $(0,1), (1,2), (2,3), (3,\infty)$)	8
$k = 1$	3 (one in $(0,1)$, $x = 2$, one in $(3,\infty)$)	6
$1 < k < 3$	2 (one in $(0,1)$, one in $(3,\infty)$)	4
$k = 3$	1 (in $(3,\infty)$, namely $x = 4$)	3 (including $x = 0$)
$k > 3$	1 (in $(3,\infty)$)	2

24a. Suppose (a,b) is on both graphs: $f(a) = b$ and $f^{-1}(a) = b$, so $f(b) = a$. If $a < b$, then since f is strictly increasing, $f(a) < f(b)$, i.e. $b < a$: contradiction. If $a > b$, then $f(a) > f(b)$, i.e. $b > a$: contradiction. Hence $a = b$.

24b. $f(x) = 1 - x$. Then $f^{-1}(x) = 1 - x = f(x)$, so the graphs coincide and meet for example at $(0,1)$, which is not on $y = x$. (Another example: $f(x) = -x^3$ and $f^{-1}(x) = -\sqrt[3]{x}$ meet at $(1,-1)$.)

25a. Write $A = f(x)$ and $B = f\left(\frac{1}{x}\right)$. The given equation is $A + 2B = 3x$. Replacing x by $\frac{1}{x}$ in the given equation gives $f\left(\frac{1}{x}\right) + 2f(x) = \frac{3}{x}$, i.e. $2A + B = \frac{3}{x}$. Multiply the second by 2: $4A + 2B = \frac{6}{x}$. Subtract the first: $3A = \frac{6}{x} - 3x$, so

$$f(x) = \frac{2}{x} - x.$$

Check: $f(x) + 2f\left(\frac{1}{x}\right) = \frac{2}{x} - x + 2\left(2x - \frac{1}{x}\right) = \frac{2}{x} - x + 4x - \frac{2}{x} = 3x$.

25b. $f(-x) = -\frac{2}{x} + x = -f(x)$, so f is odd. Then $f(x) = f(-x) \iff f(x) = -f(x) \iff f(x) = 0 \iff \frac{2}{x} = x \iff x^2 = 2 \iff x = \pm\sqrt{2}$.